

Pranav Gopal Singh Thakur

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Professional Summary: Mechanical & Manufacturing candidate who has worked in the industrial and operational segments, orchestrating leadership, managing 30-member teams, and achieving high performance metrics, \$120K revenue growth.

EDUCATION

Duke University

Durham | NC | USA

Master of Science in Mechanical Engineering & Material Science (GPA: 3.85/4.0)

Aug 2025 – Dec 2026

Minor Certificate: Innovation and Entrepreneurship

Relevant Coursework: Aerodynamics, FEM, Business & Org Fundamentals, Supply Chain Management, Experimental Design

Teaching Assistant: ME321 Strength of Materials & SolidWorks (Prof Steven Creighton), ME344 Control Systems (Prof Gustafson)

Manipal Institute of Technology

Karnataka | India

Bachelor of Technology in Mechanical Engineering (GPA: 8.34/10.0)

Aug 2020 - Jul 2024

Minors: Manufacturing

Relevant Coursework: Mechanical Design, CAD/CAM, Quality Control, Supply Chain, Control System, Lean Manufacturing, Python, SQL, ML, Finance, Advanced Manufacturing Processes, Jet Propulsion & Rocket Technology, Thermodynamics, Nano Tech & MEMS

EXPERIENCE

Manufacturing & Industrial Engineer

Jan 2024 – Jul 2025

Balaji CNC Technologies (BCT Group)

Hyderabad | India

- Engineered and architected aircraft/rocket components; Utilized GD&T, CAD/CAM/CAE technologies for multi-assembly integration; 75 components produced, reducing cycle/setup times by 25%.
- Developed and integrated an ERP system for AS9100 certification; Employed API, Python, AWS for workflow management across teams; Reduced supply chain lead time by 15-20%.
- Forecasted and scheduled production capacity; Leveraged Power BI to enhance operational efficiency; Achieved \$120K annual revenue growth, boosting customer satisfaction scores.
- Restructured OEE metrics; Established KPI improvements leading to new customer acquisition; Secured relationships with 4 new customers, contributing to a 60% growth in fiscal year.
- Optimized inventory management; Implemented ABC analysis and EOQ strategies, visualized using Tableau for procurement; Reduced tool procurement costs by 10%, ensuring seamless production flow.

INTERNSHIPS & PROJECTS

Project: FSAE, Duke Motorsports | Design & Aerodynamics Engineer

Duke | Aug 2025 – Jan 2026

- Conducted wind tunnel testing for design and validation; executed tests to refine NACA profile designs, resulting in 15% downforce improvement for vehicle stability during high-speed maneuvers. Achieved a skidpad lap time reduction of 16%.

Project: Altair Sounding Rocket (SAC 2023), Thrust MIT | Structures & Manufacturing Engineer

Manipal | Jan 2022 – Aug 2023

- Led system design projects, covering end-to-end sounding rocket architecture, prototyping, and performance optimization across projects. ("#1 position in Asia at the Spaceport America Cup (SAC) 2023").
- Engineered and fabricated the Altair sounding rocket using CATIA, Fusion 360, and ANSYS, cutting production lead time by 40%.
- Built a parallel-manipulator payload in a 4U CubeSat with an MPU-based control, achieving 3 DOFs.
- Devised the team's first inverted thrust stand for a solid motor that delivered the target neutral thrust curve with a 3.6 s burn.

Internship: DRDL | Mechanical Intern

Hyderabad | Dec 2022 – Jan 2023

- Modelled pressure regulator for propulsion systems using Creo, validated in MATLAB Simulink, AMESim with 98% precision.

SKILLS

- Manufacturing & Design:** SolidWorks 2025, Mastercam 2024, CATIA 2024, AutoCAD 2025, Creo 2018, Fusion 360 - 2025, Power Mill 2022, Ansys, CFD Fluent, HyperMesh, GD&T, CNC Mill/Turn, EDM, CMM, Water Jet Cutting, 3D Printing, Profile Projectors, Height Gauges, Subsonic Wind Tunnel, MATLAB, Simulink, AMESim.
- Quality Control:** FAI, PPAP/APQP, Root Cause Analysis, Statistical Process Control, Quality Control Instruments, BOM, Value Stream Mapping, Height Gauges, Six Sigma, Route Cards, SOPs.
- Data Analysis & ERP:** Python, Power BI, Tableau, SQL (MySQL, Postgres), ERP/MRP (SAP), API Integration, Workflow Management, OEE Metrics, Vendor and Supplier Management, Web Development, Minitab.
- Certifications:** Ansys Workbench Certified, SolidWorks CSWA, AS9100 Certification Experience, ISO 9001.

PUBLICATIONS & PATENT

Utkarsh Anand, Diya Parekh, Thakur Pranav G. Singh, Hrishikesh S. Yadav, Ramya S. Moorthy, Srinivas. "Design and control of "STEWIE", a robotic payload stabilization mechanism for rocket flights." (Patent Application no: 202341075776)